Kasa Telecom: Customer churn (next 90 days)

Insight report · data as of 2026-09-30 · prepared 14 October 2026 · Christokin Christopherson

4,173 Active customers	**GHS 721,925** Monthly recurring revenue	**7.22%** 90-day churn rate	**276** Expected leavers, next 90 days	**GHS 424,216** Revenue at risk (12 months)

Executive summary

AI-generated by the Snwolley agent (mock) · Verified - every figure matches the source data

Situation

Kasa Telecom serves 4,173 active customers worth GHS 721,925 in monthly recurring revenue. In the last 90 days 295 customers left (7.22% of the base) against 384 new customers.

Key findings

- Churn peaked in 2025-10 at 3.75% a month.
- Greater Accra saw the sharpest regional spike: 4.48% monthly churn from 2026-04 to 2026-06 vs 2.37% elsewhere.
- Month-to-month Prepaid customers are the most likely to leave (10.83% within 90 days).
- Falling usage is the clearest early warning: 8.65% churn vs 4.45% for stable users.
- 'Price' is the most common recorded cancellation reason (25.1%).
- Data quality: 13 of 14 checks found issues; all are corrected in the analytics layer, but high-severity ones should be fixed at source.

What the model tells us

The gradient boosting model identifies the riskiest 15% of customers who account for 54.1% of future churners (3.61x random; ROC-AUC 0.78).

It expects about 276 customers to leave in the next 90 days, with GHS 424,216 of revenue at risk over 12 months.

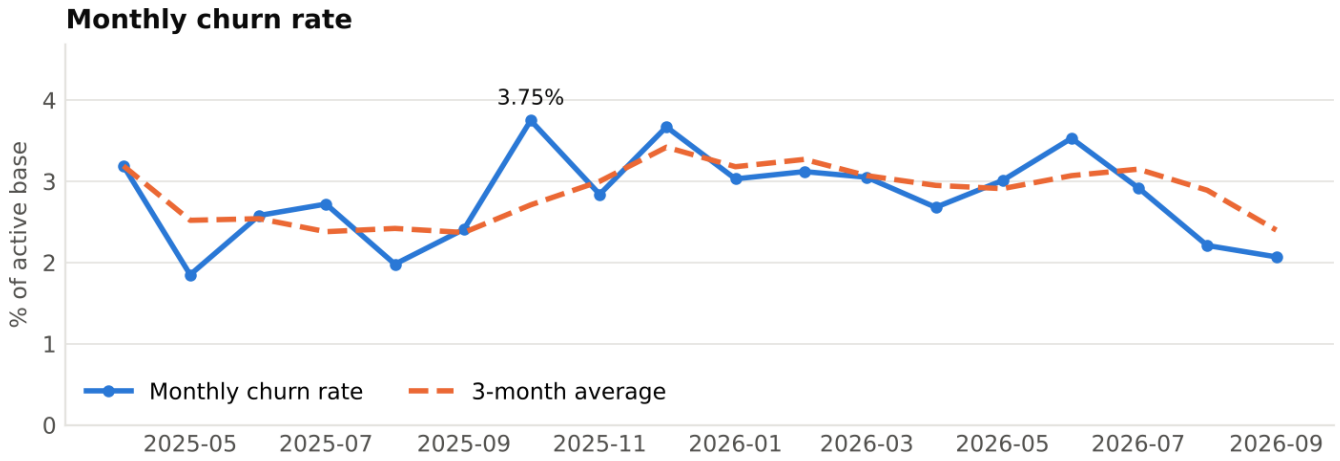
A targeted retention campaign is estimated to return GHS 40,684 net (ROI 3.42) versus GHS 4,651 if customers were picked at random.

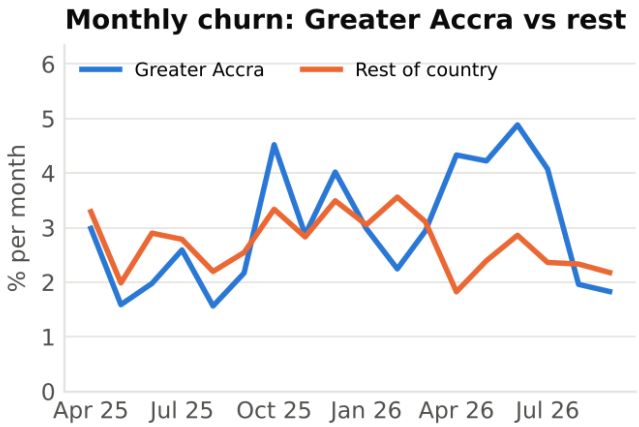
Recommendations

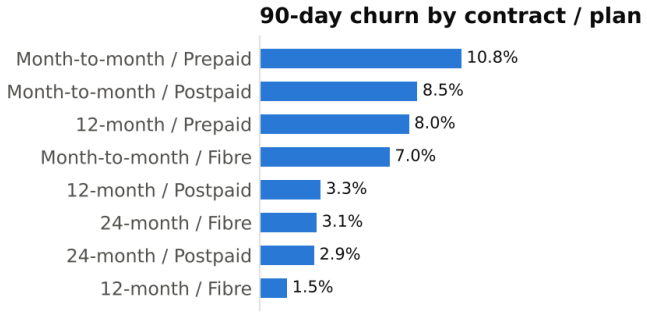
1. Act on the High-risk tier first, starting with month-to-month prepaid customers, keeping a control group to prove impact.
2. Turn the strongest early-warning driver into an automatic trigger for intervention.
3. Fix the high-severity data issues at source and re-score monthly, monitoring drift.

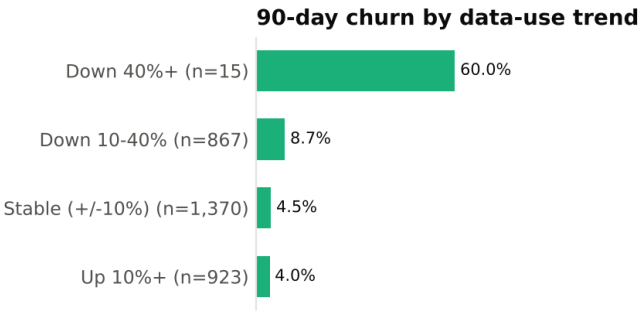
What the data shows

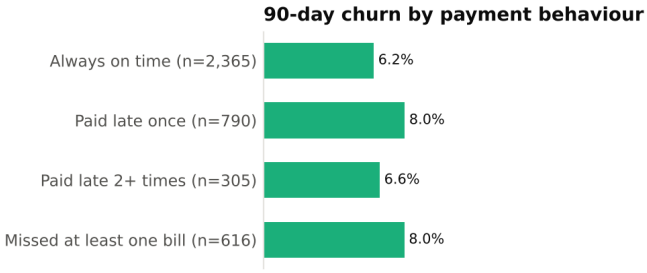
All figures below come from SQL run on the cleaned data layer. 90-day rates follow customers active at the last fully observed snapshot (2026-06-30) for 90 days.











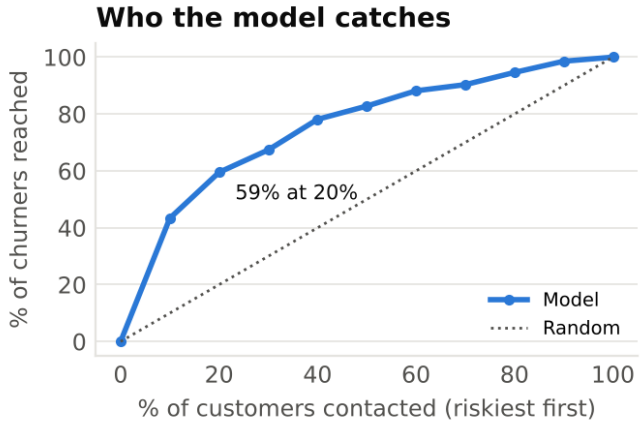
Recorded cancellation reasons (last 12 months)

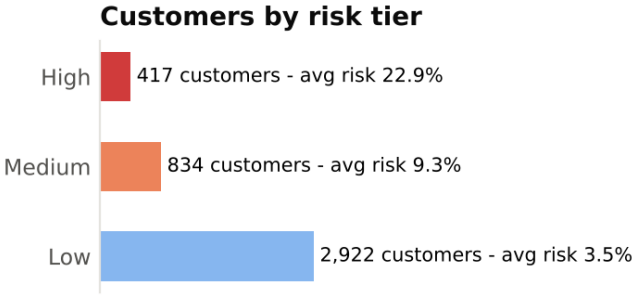
Reason	Cancellations	Share	Lost monthly revenue
Price	350	25.1%	GHS 36,282
Not captured	249	17.9%	GHS 26,650
Poor network	192	13.8%	GHS 20,165
Moved to competitor	172	12.3%	GHS 19,919
Poor customer service	170	12.2%	GHS 18,455
Non-payment	140	10.1%	GHS 24,088
Relocation	72	5.2%	GHS 8,164
Other	48	3.4%	GHS 5,372

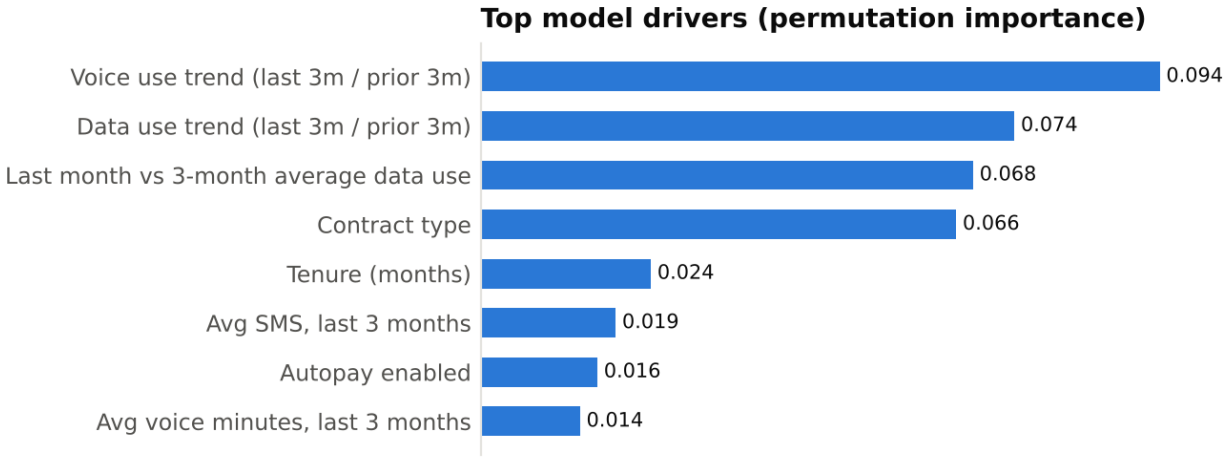
The predictive model

Outcome predicted: **Customer churn (next 90 days)**. Trained on 14,727 historical snapshots (2025-06-30, 2025-09-30, 2025-12-31, 2026-03-31) and tested on 4,076 customers at 2026-06-30 (out-of-time validation, so the test mimics real future use). Selected model: **Gradient boosting** (highest PR-AUC).

Model	ROC-AUC	PR-AUC	Reach @ top 15%	Lift	Brier
Baseline (base rate)	0.500	0.068	16.8%	1.12x	0.0640
Logistic regression	0.765	0.307	51.6%	3.44x	0.0561
Random forest	0.773	0.313	50.5%	3.37x	0.0566
Gradient boosting	0.780	0.340	54.1%	3.61x	0.0543







Drivers are measured by how much test performance drops when a feature is shuffled. They show what the model relies on, not proven causes.

Value of acting on the model (simulated on the test month)

	Model-targeted	Random selection
Customers contacted	611	611
Would-be churners reached	151	42
Customers saved (at assumed save rate)	45	13
Revenue protected	GHS 52,567	GHS 14,569
Net benefit after campaign cost	GHS 40,684	GHS 4,651

Assumptions: contact top 15% of customers; contact cost GHS 15; incentive GHS 60 per saved customer; save rate 30%; revenue valued over 12 months. Change these in the use-case configuration.

Where the predicted risk sits

Contract	Plan	Customers	High risk	Avg risk	Revenue at risk / month
Month-to-month	Prepaid	1,503	315	10.5%	GHS 10,391
12-month	Postpaid	855	18	3.9%	GHS 5,880
Month-to-month	Postpaid	422	54	7.8%	GHS 5,766
12-month	Fibre	292	1	2.9%	GHS 3,736
Month-to-month	Fibre	135	12	6.0%	GHS 3,608
24-month	Postpaid	491	6	3.2%	GHS 2,726
24-month	Fibre	206	2	2.6%	GHS 2,292
12-month	Prepaid	269	9	5.5%	GHS 953

Recommended actions

● AI-generated by the Snwolley agent (mock) · Verified - every figure matches the source data

Priority segments

The model places 417 customers in the **High** tier (average predicted risk 22.9%); about 95 of them are expected to leave. They carry GHS 102,745 of revenue at risk over 12 months.

- Month-to-month / Prepaid: 315 high-risk customers, GHS 10,391 expected monthly revenue at risk.
- 12-month / Postpaid: 18 high-risk customers, GHS 5,880 expected monthly revenue at risk.
- Month-to-month / Postpaid: 54 high-risk customers, GHS 5,766 expected monthly revenue at risk.

Regions with the most revenue at risk: Greater Accra (GHS 11,331/month), Ashanti (GHS 7,054/month), Eastern (GHS 2,705/month).

Recommended actions

1. Offer month-to-month customers a discounted 12-month plan in exchange for commitment. *(signal: Contract type - 365 high-risk customers)*
2. Trigger a re-engagement offer (bonus data) when a customer's data use drops sharply, and check network quality in their area. *(signal: Last month vs 3-month average data use - 206 high-risk customers)*
3. Run an onboarding call and a 60-day check-in for new customers. *(signal: Tenure (months) - 151 high-risk customers)*
4. Include voice bundles in the re-engagement offer for customers whose calling has dropped. *(signal: Voice use trend (last 3m / prior 3m) - 138 high-risk customers)*

How to measure success

- Hold out a random control group from the High tier to measure the true save rate against the planned 30%.
- Track 90-day churn of treated versus control customers, cost per save (planned contact cost GHS 15) and revenue retained.
- Re-score monthly and compare predicted with actual churn to confirm the model is still accurate.

Data quality

Automated SQL checks on the raw data. Every issue is handled in the cleaned layer before analysis.

Check	Severity	Records	%	Handling
Customers with no region recorded	medium	266	3.28	Kept; region treated as "Unknown" category in the model.
Customers with no date of birth	low	281	3.46	Age imputed with the median inside the ML pipeline; a missing-flag is added.
Region labels that do not match the reference list (spelling / case / city names)	medium	770	9.48	Standardised in v_customers_clean (e.g. 'Gt. Accra', 'kumasi' -> canonical region).
Duplicate customer registrations (same phone number, different customer id)	high	120	1.48	De-duplicated in v_customers_clean - earliest record kept (ROW_NUMBER window).
Subscriptions whose end date is before the start date	high	15	0.19	Quarantined (excluded) in v_subscriptions_clean; needs correction in the source system.
Subscriptions whose status disagrees with the end date (e.g. 'active' but cancelled)	high	22	0.28	Status re-derived from end_date in v_subscriptions_clean.
Duplicate usage records for the same subscription and month	medium	1,129	0.79	De-duplicated in v_usage_clean (first record kept).
Usage readings that are negative or missing	medium	1,090	0.77	Negative values set to NULL; missing values imputed in the ML pipeline.
Usage readings above 5x the plan allowance (meter / mediation errors)	low	131	0.09	Set to NULL in v_usage_clean.
Bills posted more than once for the same subscription and month	high	706	0.50	De-duplicated in v_payments_clean - otherwise revenue would be overstated.
Payments dated before the billing month, or paid amount > 3x the amount due	medium	448	0.32	Flagged (has_inconsistency) and overpayments capped at amount due in v_payments_clean.
Support tickets with CSAT outside the valid 1-5 scale	low	172	0.57	Set to NULL in v_tickets_clean.
Support tickets with a negative resolution time	low	85	0.28	Set to NULL in v_tickets_clean.

Method, assumptions and limitations

- Data: synthetic but realistic operator data generated by a reproducible script (fixed seed); no real customer is represented.
- Features are extracted with SQL using only information available at each snapshot date, preventing leakage of future information.
- Model choice: four candidates including a no-skill baseline; selected on PR-AUC because the outcome is rare.
- AI narrative: produced by the Snwolley agent from a curated, de-identified context pack; every figure is checked against the source data and the result is labelled.
- Monitoring: input and score drift are measured with the Population Stability Index each time customers are re-scored.
- Limitations: business value depends on the stated campaign assumptions; drivers are associations, not causes; a controlled test is required before scaling.